Cross-Currency Dynamics in Cryptocurrencies

Abstract. When Silicon Valley Bank collapsed in March 2023, USDC traded as low as \$0.87, fragmenting Bitcoin prices across exchanges and quote currencies. Using 1-minute data from Binance, Coinbase, and Kraken, we decompose this fragmentation into a large stablecoin-driven component (USDC crisis-mean dispersion $D_t \approx +320$ bps) and a smaller adjusted residual ($B_t \approx +5$ bps, half-lives 0.54–1.10 minutes). A two-layer dynamic emerges: the adjusted basis mean-reverts in $\approx$0.6 minutes during crisis, while the USDC peg deviation half-life extends to $\approx$572 minutes—a $\sim$940$\times$ gap confirmed by sieve bootstrap ($p<10^{-4}$). A contagion intensity model formalizes this linkage: the transmission parameter $\hat{\lambda}$ is significant during crisis ($p<0.001$) but not pre-SVB, indicating that peg stress propagates to the adjusted basis beyond the mechanical price channel. USDC and USDT transmit in opposite directions, consistent with de-peg versus safe-haven dynamics. USDC crisis excess kurtosis reaches 12.9 (6$\times$ P99 tail expansion); USDT trades at a $+58$ bps safe-haven premium; and cross-stablecoin correlation drops from 0.44 to 0.16. Fiat-quoted BTC leads price discovery (Hasbrouck IS 0.73), while post-friction arbitrage opportunities narrow sharply. These findings map directly to the GENIUS Act's reserve-composition, timely-redemption, transparency, and bankruptcy provisions.

1 Introduction

Stablecoins are the primary quote currency on most crypto exchanges. Because they are not identical to bank deposits—redeemability, custody, and transparency differ—their prices can deviate from \$1, introducing a *stablecoin funding risk premium*: BTC trades at different nominal prices across quote currencies purely because of stablecoin-specific risk. Prior work on cross-exchange price dispersion has focused on fiat-quoted arbitrage and market efficiency, while studies of stablecoin de-pegs have largely examined daily price and market-cap dynamics. Less understood is how a sudden loss of reserve access propagates through high-frequency cross-currency bases, liquidity, and price discovery—the microstructure channel through which peg risk becomes trading risk. In 2025, the U.S. enacted the GENIUS Act, establishing a federal framework for dollar-pegged stablecoins with stricter reserve backing and transparency [11].

We analyze March 1–21, 2023 (UTC), spanning the SVB failure. When SVB collapsed on March 10, $\approx$\$3.3B of Circle's USDC reserves became inaccessible, and USDC de-pegged to \$0.874 on Kraken [3]. Figure 1 shows USDC falling sharply below par while USDT trades at a premium—consistent with stablecoin substitution under stress. This episode provides a high-frequency stress test of cross-currency basis behavior, liquidity fragmentation, and price discovery.

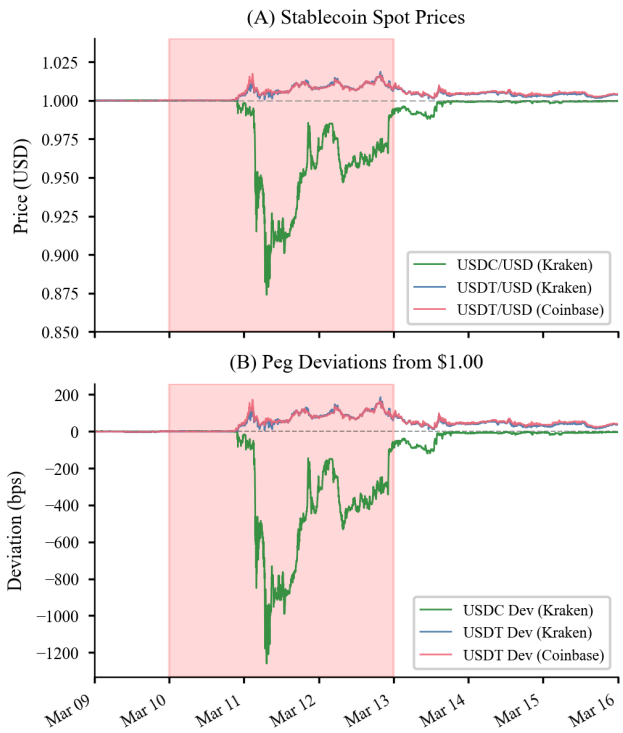

Figure 1: Stablecoin prices (A) and peg deviations (bps, B), March 2023. USDC falls to \$0.874; USDT trades at a premium. Shaded: SVB crisis (March 10–13, UTC).

We make four contributions. First, we decompose stablecoin-driven fragmentation into a peg-deviation component and an adjusted residual, re-

vealing a $\sim 940\times$ half-life gap—statistically confirmed via sieve bootstrap ($p<10^{-4}$)—that localizes persistent dislocation to the reserve-access layer. Second, we estimate a contagion intensity model showing that peg stress transmits to the adjusted basis beyond the mechanical price channel ($\hat{\lambda}$ significant at $p<0.001$ during crisis, insignificant pre-SVB), formalizing the two-layer linkage as a testable transmission parameter. Third, we document pronounced asymmetry between USDC and USDT—in tail risk, volatility, contagion direction, and cross-stablecoin correlation—that is inconsistent with a common-shock explanation. Fourth, we map these empirical findings to specific provisions of the GENIUS Act, providing a quantitative lens for evaluating stablecoin regulation.

The remainder of this paper is organized as follows. Section 2 describes the data and methodology. Section 3 presents the empirical results across five dimensions: dispersion decomposition, cross-exchange dispersion, market microstructure, price discovery, and arbitrage. Section 4 interprets the findings through a regulatory lens, and Section 5 concludes.

2 Data and Methodology

2.1 Data sources and preprocessing

We collect 1-minute OHLCV candles for BTC quoted in USD, USDT, and USDC from Binance (BTC/USDT, BTC/USDC; dominant USDT volume), Coinbase (BTC/USD, BTC/USDT, USDT/USD; primary U.S. fiat gateway), and Kraken (BTC/USD, BTC/USDT, BTC/USDC, USDT/USD, USDC/USD; broadest stablecoin-to-fiat coverage). Stablecoin-to-USD conversion uses venue-specific legs. All series are aligned to a 1-minute UTC grid (Table 1). Coverage exceeds 97% for most series, though pre-SVB Kraken BTC/USDC falls to 89.6% with 42% forward-fill share. Short gaps ($\leq$ 5 min) are forward-filled; larger gaps remain NaN. Core directional signs are stable under strict no-FF filtering (crisis USDC $D_t > 0$, USDT premium > 0), though some magnitudes shift on thinner samples. We interpret high-frequency levels as directional unless no-FF retention is adequate.

Table 1: 1-minute data coverage and forward-fill exposure on the unified UTC grid. Coverage = non-missing share; FF share = carry-forward share (max 5 minutes).

Pair	Coverage Overall (%)	Coverage Pre-SVB (%)	Coverage Crisis (%)	Coverage Post-SVB (%)	FF Share (%)
Kraken BTC/USD	99.98	100.00	100.00	99.95	0.18
Kraken BTC/USDT	98.44	97.36	99.81	99.06	30.84
Kraken BTC/USDC	94.05	89.62	99.79	96.57	42.07
Kraken USDC/USD	97.80	96.03	99.35	99.06	28.57
Kraken USDT/USD	99.98	100.00	100.00	99.95	0.91
Coinbase BTC/USD	99.10	97.90	100.00	100.00	0.06
Coinbase BTC/USDT	98.93	97.53	99.95	99.98	17.98
Coinbase USDT/USD	99.02	97.72	100.00	100.00	0.06
Binance BTC/USDT	100.00	100.00	100.00	100.00	0.00
Binance BTC/USDC	100.00	100.00	100.00	100.00	0.00

We proxy transaction frictions using the intra-candle range:

$$\text{Intra-min Range (bps)} = \left(\frac{\text{High}_t - \text{Low}_t}{\text{Close}_t}\right) \times 10000. \tag{1}$$

This is a coarse range proxy that mechanically loads on volatility and microstructure noise; we interpret it cautiously as a friction indicator, not a quoted bid–ask spread.

2.2 Core quantitative objects

For stablecoin $Q \in \{\text{USDC}, \text{USDT}\}$, we track two objects. The *unadjusted dispersion*:

$$D_{Q,t} = \big(\ln(P_{\text{BTC}/Q,t}) - \ln(P_{\text{BTC/USD},t})\big) \times 10000, \tag{2}$$

and the *adjusted parity residual*:

$$B_{Q,t} = \big(\ln(P_{\text{BTC}/Q,t} \times P_{Q/\text{USD},t}) - \ln(P_{\text{BTC/USD},t})\big) \times 10000. \tag{3}$$

These satisfy $B_{Q,t} = D_{Q,t} + 10000 \cdot \ln(P_{Q/\text{USD},t})$: large de-pegs mechanically drive D_t while B_t captures the residual after marking to USD. Both are computed intra-exchange; cross-venue deviations are studied separately. Basis and dispersion are in log-bps; peg deviations, ranges, and volatility summaries are in arithmetic bps.

2.3 Mean-reversion dynamics

We model B_t as an OU process $dX_t = \theta(\mu - X_t)dt + \sigma dW_t$ and estimate the discrete AR(1) form $X_t = c + \rho X_{t-1} + \varepsilon_t$. Stationarity is checked with ADF tests (with large N, rejection is expected and serves as a baseline screen). The half-life uses the exact mapping:

$$\tau_{1/2} = \frac{\ln(2)\,\Delta t}{-\ln(\rho)}, \quad 0 < \rho < 1. \tag{4}$$

If $\rho \leq 0$ or $\rho \geq 1$, half-life is undefined (NaN). The OU framework assumes Gaussian innovations; crisis-period B_t exhibits substantial excess kurtosis (Section 3.3), so we use the AR(1) model solely

for its half-life mapping, not as a full distributional model. We report robustness at 5-minute sampling and with forward-filled minutes removed.

2.4 Contagion intensity model

The univariate OU model treats the adjusted basis B_t in isolation from the peg layer. To formalize the inter-layer linkage, we extend the mean-reversion equation to include lagged peg deviation S_t (in bps) as a forcing variable:

$$\Delta B_t = a + \beta\, B_{t-1} + \lambda\, S_{t-1} + \varepsilon_t, \qquad (5)$$

where the *contagion intensity* λ measures the marginal rate at which peg-layer stress transmits to the cross-currency basis after stablecoin-to-USD adjustment. When peg risk is fully absorbed by the conversion leg, $\lambda{=}0$ and B_t follows a standard AR(1). Rejection of $\lambda{=}0$ indicates residual transmission—via liquidity withdrawal, adverse selection, or order-book asymmetry—beyond the mechanical price channel. We estimate (5) by OLS with HAC standard errors (Newey–West, 60 lags) [9] separately for each regime, so that both the mean-reversion speed $\hat{\rho}{=}1{+}\hat{\beta}$ and the contagion parameter $\hat{\lambda}$ are regime-specific. The lagged structure avoids simultaneity: reverse causation from B_t to the peg price is implausible at the 1-minute horizon because arbitrageurs correcting B_t do not move the aggregate stablecoin/USD rate.

2.5 Price discovery framework

For the two primary Kraken channels (BTC/USD vs. BTC/USDC and BTC/USD vs. BTC/USDT), we use synchronized 1-minute log-price levels, dropping forward-filled minutes. We test for cointegration with Johansen trace tests (constant restricted to the cointegration relation; lag order via BIC on the levels VAR) [8, 4]. When rank $= 1$, we estimate:

$$\Delta \mathbf{p}_t = \alpha\beta^\top \mathbf{p}_{t-1} + \sum_{i=1}^{k_\Delta} \Gamma_i \Delta \mathbf{p}_{t-i} + \mathbf{u}_t. \qquad (6)$$

The market with smaller $|\alpha|$ adjusts less and is interpreted as the relative leader. We retain BH/FDR-corrected Granger tests on returns as secondary evidence only [6]; with 1-minute bars, asynchrony and large N can overstate economic lead–lag strength.

3 Empirical Results

3.1 Dispersion, adjusted residual, and mean reversion

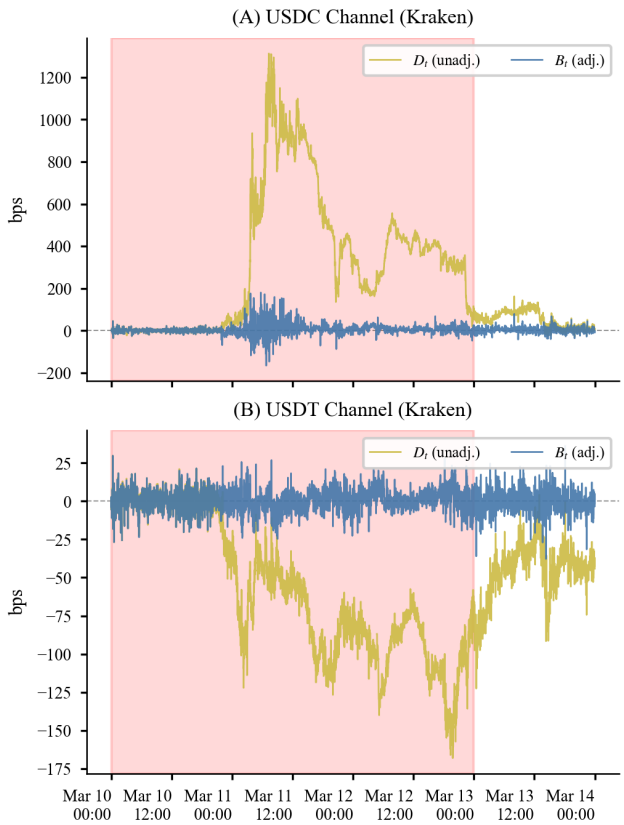

Figure 2: Kraken crisis decomposition: unadjusted dispersion D_t vs. adjusted residual B_t. USDC D_t exceeds $+1{,}200$ bps while B_t stays within ± 150 bps; for USDT, D_t reflects the premium and $B_t \approx 0$.

Before SVB, both D_t and B_t hover near zero. During the crisis (March 10–13 UTC), USDC dispersion jumps to $+320$ bps ($\sigma{=}323$) while the adjusted residual remains small ($+5.3$ bps, $\sigma{=}21$). USDT moves oppositely: $D_t{=}{-}57$ bps, $B_t{=}{+}0.7$ bps (Table 2). Figure 2 visualizes this: D_t (orange) captures the peg shift while B_t (blue) remains tightly mean-reverting.

Two dynamics drive this pattern. USDC trades as low as \$0.874 on Kraken on March 11 ($\approx{-}1{,}260$ bps), mechanically pulling D_t upward; simultaneously, USDT trades at a premium averaging $+58.1$ bps on Kraken and $+59.7$ on Coinbase. Figure 3 visualizes this stablecoin substitution: crisis observations (red) migrate to the lower-right quadrant ($D_{\text{USDC}} > 0$, $D_{\text{USDT}} < 0$), and the unadjusted cross-stablecoin correlation flips from $+0.44$

to -0.42—consistent with traders rotating out of USDC into USDT under stress.

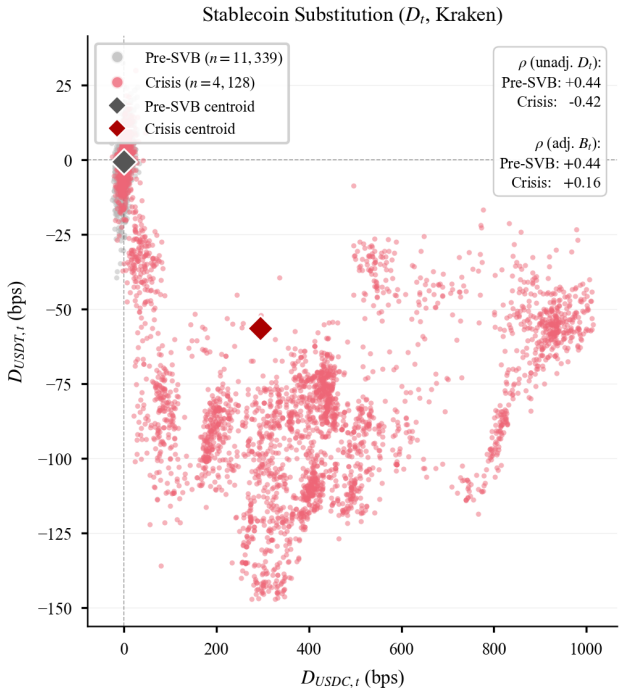

Figure 3: Kraken stablecoin substitution: $D_{\text{USDC},t}$ vs. $D_{\text{USDT},t}$. Crisis points shift to the lower-right quadrant; unadjusted correlation flips ($+0.44 \rightarrow -0.42$) and adjusted B_t correlation falls ($0.44 \rightarrow 0.16$).

Table 2: Kraken regime statistics for unadjusted dispersion (D_t) and adjusted residual (B_t), in bps.

Regime	Series	Mean (bps)	Std (bps)	Mean Abs (bps)	N
Pre-SVB	USDC D_t (Unadj.)	0.12	4.95	3.49	11615
Pre-SVB	USDC B_t (Adj.)	-0.32	5.00	3.53	11197
Pre-SVB	USDT D_t (Unadj.)	-0.54	4.49	3.17	12618
Pre-SVB	USDT B_t (Adj.)	-0.60	4.33	3.02	12618
Crisis	USDC D_t (Unadj.)	319.97	322.87	321.17	4311
Crisis	USDC B_t (Adj.)	5.27	21.21	13.37	4283
Crisis	USDT D_t (Unadj.)	-57.18	45.10	58.57	4312
Crisis	USDT B_t (Adj.)	0.68	6.57	4.94	4312
Post-SVB	USDC D_t (Unadj.)	11.22	21.95	14.27	12514
Post-SVB	USDC B_t (Adj.)	0.45	9.06	6.68	12419
Post-SVB	USDT D_t (Unadj.)	-30.26	13.29	30.28	12837
Post-SVB	USDT B_t (Adj.)	-1.23	6.47	4.92	12837

HAC/Newey–West intervals (60-lag cap) confirm directional robustness [9]: USDC crisis D_t 95% CI [245.7, 394.3] bps; crisis B_t CI [3.6, 7.0] bps; USDT premiums positive on both Kraken ([47.4, 68.8]) and Coinbase ([49.1, 70.3]). Coinbase–Kraken BTC/USD remains significantly positive ([1.7, 2.6] bps).

ADF rejects a unit root for all B_t channels ($p < 0.001$). Kraken USDC half-life: 0.98 min pre-SVB, 0.61 min crisis, 0.79 min post-SVB. Figure 4 shows robustness across 1m/5m sampling and all/no-FF filters: USDC crisis half-life ranges from 0.54 to 1.10 min; USDT crisis half-life is longer (0.92–3.06 min). Fast USDC mean reversion is robust across all specifications.

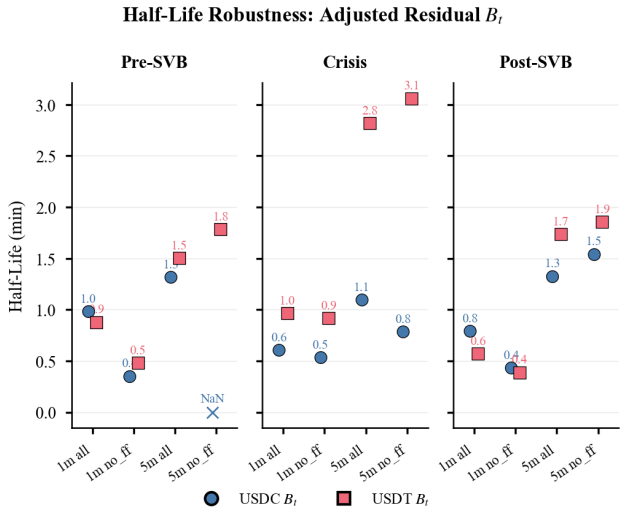

Figure 4: Adjusted-residual B_t half-life robustness by regime/specification (1m/5m $\times$ all/no-FF). USDC crisis half-life stays below 1.1 min; USDT reaches 3.1 min (5m no-FF).

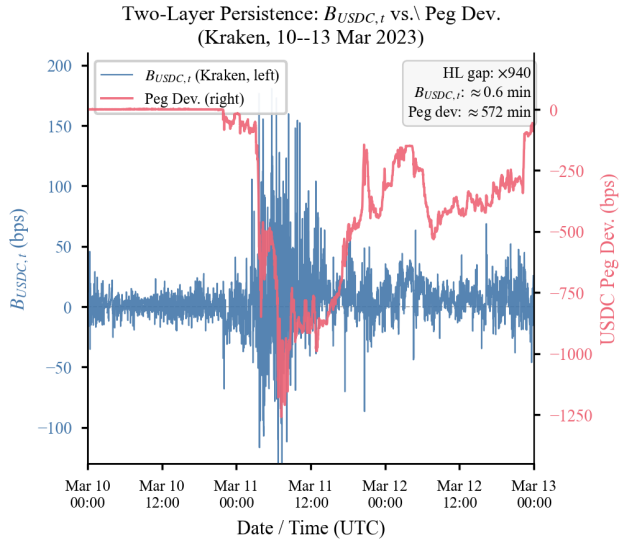

Figure 5: Two-layer persistence (March 10–13): $B_{\text{USDC},t}$ mean-reverts in $\approx$0.6 min, while the USDC peg deviation persists for $\approx$572 min. The $\sim$940$\times$ gap localizes persistent dislocation to the peg layer.

The OU estimation reveals a **two-layer dynamic** (Figure 5): B_t mean-reverts in $\approx$0.6 min during crisis ($\hat{\rho}$=0.320), while the USDC peg deviation persists with half-life $\approx$572 min ($\hat{\rho}$=0.999, ADF p=0.44); USDT peg deviations similarly persist (496–542 min). The point-estimate ratio $R = \tau^{\text{peg}}_{1/2}/\tau^{B}_{1/2} \approx 940$ is striking, but could in principle reflect estimation noise. A parametric sieve

bootstrap (10,000 replications; residuals resampled from each AR(1) to preserve near-unit-root persistence of S_t) yields $R_{\text{median}}=567$ with 95% CI [216, 1,898] and $p<10^{-4}$ for H_0: $R\leq 1$; a log-scale delta method confirms the result ($z=10.5$, $p<10^{-16}$). The persistence gap is statistically overwhelming: cross-market correction is fast even in crisis, but the peg itself—driven by reserve-access risk—is where dislocation persists.

Contagion intensity. Whether the two layers interact beyond the mechanical decomposition identity is an empirical question. Estimating the coupled model (5), we find that the contagion intensity $\hat{\lambda}$ is indistinguishable from zero pre-SVB (-0.002, $p=0.98$) but becomes significantly negative during the crisis (-0.009, $p<0.001$) and remains elevated post-SVB (-0.019, $p=0.006$); see Table 3. Because $S_t<0$ corresponds to a peg discount (USDC below par), the negative $\hat{\lambda}$ implies that a deeper peg discount at $t-1$ pushes B_t upward at t—consistent with liquidity withdrawal from the USDC book creating transient positive residual basis even after marking to USD. USDT displays the mirror pattern: $\hat{\lambda}=+0.019$ during crisis ($p<0.001$), so a USDT premium at $t-1$ pulls B_t upward—consistent with safe-haven demand spilling into the adjusted residual. These sign patterns confirm that peg stress does not merely drive D_t mechanically but also transmits to B_t through microstructure channels, and that the two stablecoins propagate stress in opposite directions.

Table 3: Contagion intensity $\hat{\lambda}$ from Eq. (5): regime-specific OLS with Newey–West HAC (60 lags). S_t is the peg deviation (bps; negative = discount).

Channel	Regime	$\hat{\lambda}$	SE	t-stat	p-value	R^2	N
USDC	Pre-SVB	−0.002	0.091	−0.02	0.985	0.253	11,196
USDC	Crisis	−0.009	0.002	−4.77	<0.001	0.351	4,282
USDC	Post-SVB	−0.019	0.007	−2.73	0.006	0.293	12,418
USDT	Pre-SVB	−0.076	0.032	−2.38	0.017	0.273	12,617
USDT	Crisis	+0.019	0.004	+5.23	<0.001	0.272	4,311
USDT	Post-SVB	+0.023	0.008	+2.80	0.005	0.354	12,836

Robustness is strong: the USDC crisis $\hat{\lambda}$ is significant under no-forward-fill sampling (-0.009, $p<0.001$, $N=3{,}001$) and at 5-minute frequency (-0.012, $p<0.001$, $N=861$). For USDT, the sign flip between pre-SVB ($\hat{\lambda}<0$) and crisis ($\hat{\lambda}>0$) coincides precisely with the shift from normal co-movement to safe-haven dynamics, consistent with the correlation break documented below.

Figure 6 provides a granular minute-by-minute view of all channels during the peak crisis window. Panel A reveals that the USDC adjusted residual (green) exhibits extreme volatility—spikes exceeding $+150$ bps—while the USDT channels on both Kraken and Coinbase remain tightly bounded near zero. Panel B shows that cross-exchange differentials widen simultaneously, with the Coinbase–Kraken BTC/USD basis rising to $+60$ bps, suggesting stress propagation across venues.

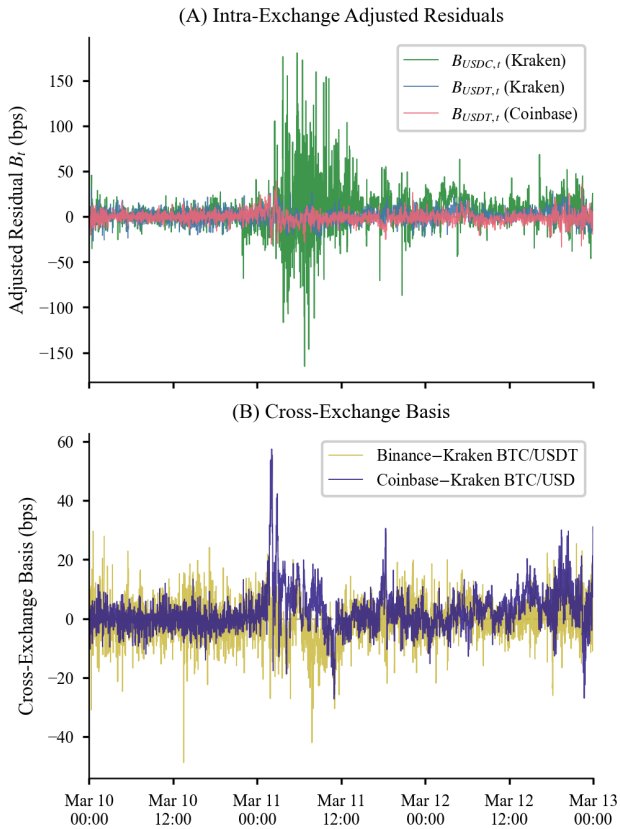

Figure 6: Crisis zoom (March 10–13). (A) $B_{\text{USDC},t}$ spikes past ± 150 bps while USDT stays within ± 30 bps. (B) Cross-exchange basis widens simultaneously, indicating stress propagation across venues.

3.2 Cross-exchange price dispersion

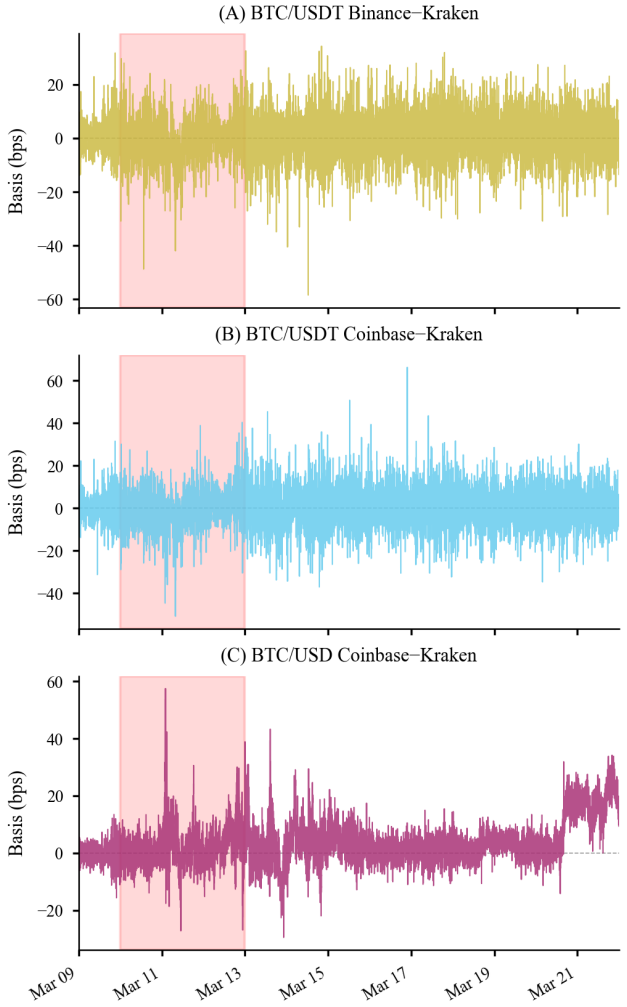

Figure 7: Cross-exchange basis (bps): BTC/USDT Binance–Kraken (A), Coinbase–Kraken (B), and BTC/USD Coinbase–Kraken (C). The fiat channel stays persistently positive and elevated post-crisis.

Having established that intra-exchange fragmentation is dominated by USDC peg risk, a natural question is whether this dislocation propagates *across* exchanges. We examine this by computing log-price differentials for BTC/USDT (Binance vs. Kraken; Coinbase vs. Kraken) and BTC/USD (Coinbase vs. Kraken). Caveat: USDT cross-exchange dispersion is in the USDT numeraire; a USD-based arbitrage requires an additional USDT/USD leg. Binance–Kraken BTC/USDT is near zero full-sample ($+0.007$ bps), consistent with tight integration. Coinbase–Kraken BTC/USD shows a persistent $+2.18$ bps premium, consistent with a structural U.S. fiat gateway effect.

During the crisis, cross-exchange differentials widen alongside intra-exchange deviations (Figure 7, Table 9), consistent with broad stress rather than a single-venue disruption. The Coinbase–Kraken fiat premium did not fully normalize post-crisis, suggesting it partly reflects structural microstructure differences that stress amplified.

3.3 Market microstructure dynamics

The cross-exchange widening documented above raises the question of whether liquidity conditions differ materially across channels. We quantify liquidity with the Roll (1984) effective spread [10] and Amihud (2002) ILLIQ ratio [1]. Table 4 and Figure 8 report regime means. Kraken BTC/USDC Roll spread jumps $23\times$ (1.10 to 25.64 bps), while BTC/USDT holds near 3 bps. Amihud confirms Coinbase BTC/USD is deepest and Kraken BTC/USDC least liquid. The two measures diverge in crisis: Roll widens while Amihud declines as volume surges.

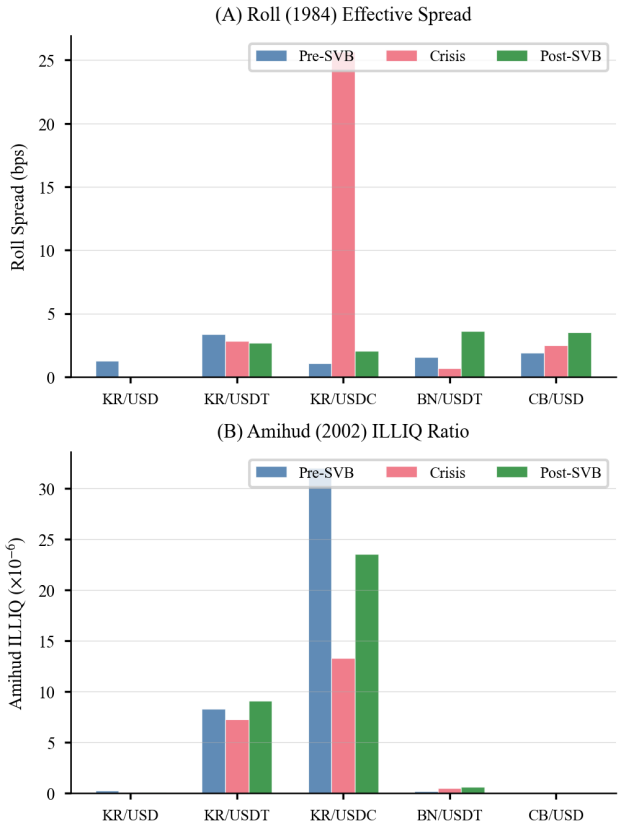

Figure 8: Roll effective spread (A) and Amihud ILLIQ (B) by pair/regime. Kraken BTC/USDC shows a $23\times$ Roll spread blowout in crisis. Roll is undefined when serial covariance is non-negative; thin-pair means are indicative.

Table 4: Roll effective spread (bps) and Amihud illiquidity ($\times 10^{-6}$) by pair/regime. Roll uses daily serial covariance of 1-minute log returns (NaN days excluded; N = valid Roll days). $\text{ILLIQ}_t = |r_t|/\text{DollarVol}_t$ (daily average).

Pair	Roll_{Pre}	$\text{Roll}_{\text{Crisis}}$	$\text{Roll}_{\text{Post}}$	N_{Pre}	N_{Crisis}	N_{Post}	$\text{ILLIQ}_{\text{Pre}}$	$\text{ILLIQ}_{\text{Crisis}}$	$\text{ILLIQ}_{\text{Post}}$
Kraken BTC/USD	1.260	—	—	4	0	0	0.232	0.050	0.068
Kraken BTC/USDT	3.390	2.850	2.670	1	1	1	8.299	7.266	9.074
Kraken BTC/USDC	1.100	25.640	2.070	2	2	2	31.970	13.282	23.530
Binance BTC/USDT	1.560	0.670	3.630	6	1	5	0.163	0.461	0.591
Coinbase BTC/USD	1.880	2.490	3.490	5	1	1	0.005	0.003	0.003

Daily dollar-volume depth proxies (Table 5) sharpen this picture: Coinbase BTC/USD is the deepest channel ($239M pre-SVB, rising to $664M post-SVB), while Kraken BTC/USDC is 26× thinner in crisis ($15.7M vs. $407M). The depth gap constrains the scale at which arbitrageurs can trade the USDC dislocation, consistent with the limited post-friction profitability documented in Section 3.5.

Table 5: Daily traded dollar volume by pair/regime (median, $MM/day). Dollar volume serves as a depth proxy where L2 order-book data is unavailable.

Pair	Pre-SVB	Crisis	Post-SVB
Kraken BTC/USD	59.50	160.36	203.77
Kraken BTC/USDT	5.55	13.84	17.37
Kraken BTC/USDC	2.56	15.65	5.52
Binance BTC/USDT	52.44	73.92	71.46
Coinbase BTC/USD	239.07	407.27	664.46

HAC regressions (Newey–West, 60 lags) of B_t on a crisis dummy, realized volatility, and range proxy (Table 6) show a significant USDC crisis coefficient (+4.30 bps, $p < 0.001$) [9]. Explanatory power is modest ($R^2 = 0.041$), so we interpret these as partial associations.

Table 6: HAC regressions of Kraken adjusted residual B_t on crisis, realized volatility, and range proxy (Newey–West, 60 lags).

	USDC Channel		USDT Channel	
	Coef.	p-value	Coef.	p-value
Constant	−0.079	0.833	−0.656	< 0.001
Crisis	+4.295	< 0.001	+1.655	< 0.001
RealizedVol (60m)	+0.004	0.928	−0.029	0.179
Range Proxy	+0.048	0.005	−0.013	0.082
R^2	0.041		0.012	
N	26,489		26,489	

OLS with HAC standard errors (Newey–West, 60 lags). Dependent variable: B_t (bps).

Figure 9 shows daily volume fragmentation: USDC share doubles during crisis (2.1% to 4.3%) as activity concentrates in Kraken BTC/USDC, while post-crisis USD dominance returns (86.9%).

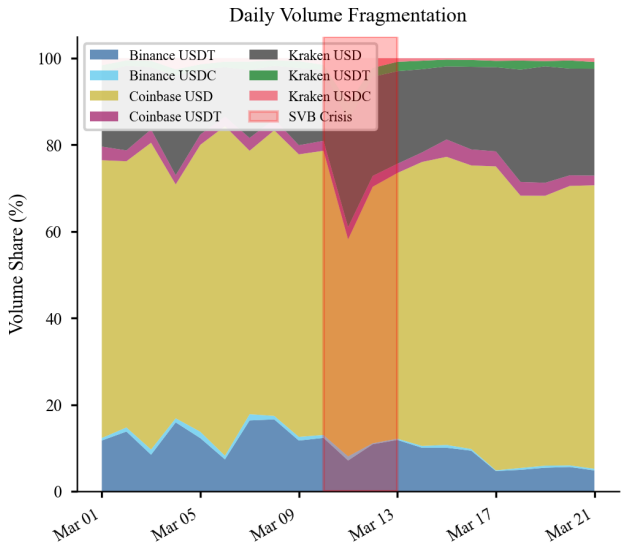

Figure 9: Daily volume shares across pairs. USDC share spikes during the crisis (Kraken BTC/USDC surge) and rotates back to USD post-crisis.

Realized volatility (Figure 10) reveals pronounced asymmetry: Kraken BTC/USDC peaks at ≈650 bps/hr on March 11, roughly 2.1× the crisis-average for BTC/USD and BTC/USDT—consistent with compounding asset risk and numeraire risk.

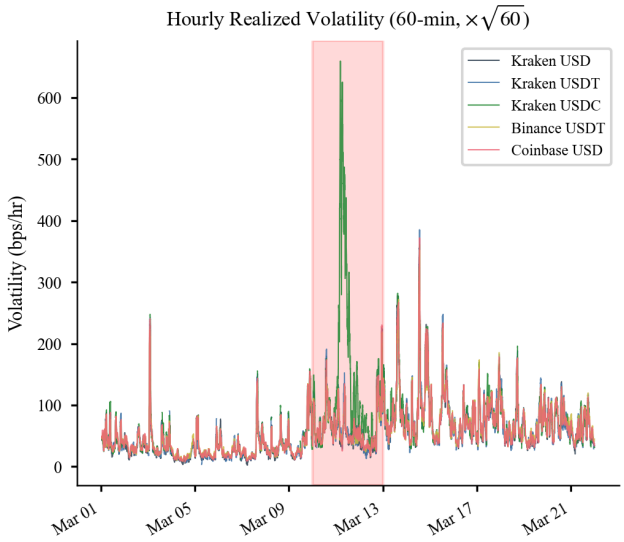

Figure 10: Hourly realized volatility (60-min rolling σ, $\times\sqrt{60}$) by pair. Kraken BTC/USDC peaks near ≈650 bps/hr in crisis, about 2.1× the USD and USDT channels.

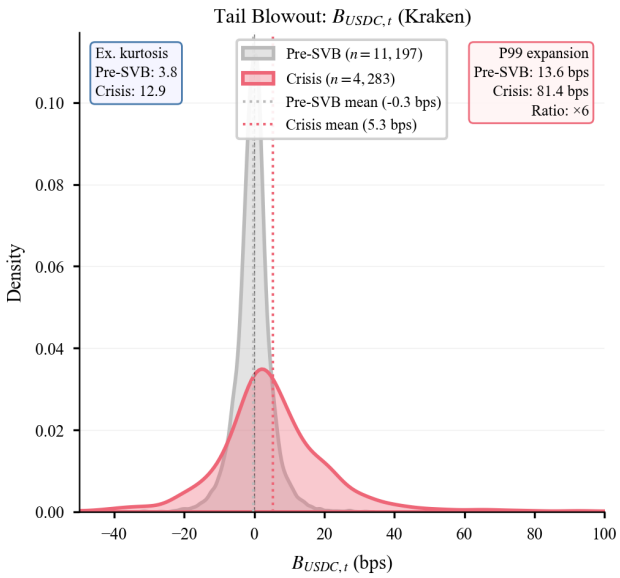

Figure 11: Kraken distribution of USDC adjusted residual $B_{\text{USDC},t}$: pre-SVB vs. crisis. Crisis shows a right shift, excess kurtosis 12.9 (vs. 3.8), and a 6× increase in the 99th percentile (13.6 to 81.4 bps).

The distributional impact is stark (Figure 11): crisis USDC B_t develops tails beyond ±100 bps versus ±10 bps pre-SVB. For risk management, stablecoin positions carry latent peg-break tail risk that calm-calibrated VaR models materially understate.

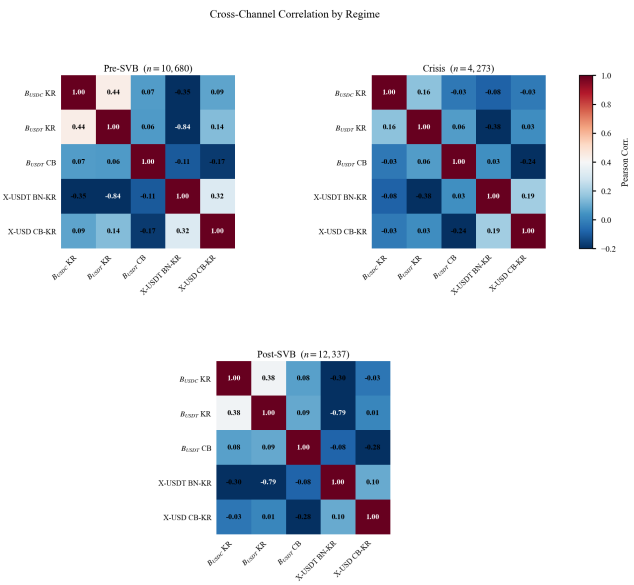

Figure 12: Regime correlation heatmap for adjusted residuals and cross-exchange basis. B_{USDC}–B_{USDT} correlation drops from 0.44 to 0.16 in crisis, indicating divergent stablecoin dynamics.

Distributional robustness. Chow tests reject parameter stability at March 10 for D_t (F=2,089), B_t (F=230), and AR(1) dynamics (F=71; all $p<10^{-16}$), validating the regime split [2]. Table 7: USDC crisis B_t has excess kurtosis 12.9 and skewness +1.01 (vs. pre-SVB 3.8/≈0); USDT crisis kurtosis is only 1.4—confirming USDC-specific tail risk. Jarque–Bera rejects normality throughout (USDC crisis JB=30,498). Cross-stablecoin correlation drops from 0.44 to 0.16 in crisis, consistent with divergent dynamics rather than a common shock.

Table 7: Higher-moment diagnostics for adjusted residual B_t by regime. Excess kurtosis is Fisher (0 under Gaussian); Min/Max in bps.

Channel	Regime	Skewness	Ex.Kurt.	Min	Max
USDC	Pre	0.00	3.8	−36.9	32.0
USDC	Crisis	1.01	12.9	−165.2	180.6
USDC	Post	0.30	2.8	−43.2	79.4
USDT	Pre	-0.36	5.2	−36.4	34.5
USDT	Crisis	-0.11	1.4	−26.8	29.9
USDT	Post	-0.03	2.0	−37.8	36.2

Figure 12 visualizes the decorrelation across all channels: pre-SVB cross-channel correlations (B_{USDC}–B_{USDT}: 0.44) collapse in crisis (0.16) and partially recover post-SVB (0.38), while cross-exchange USDT channels remain tightly correlated throughout.

3.4 Price discovery

Given the liquidity asymmetry documented above—Coinbase BTC/USD is 26× deeper than Kraken BTC/USDC—a natural question is where price discovery occurs across these fragmented channels.

Table 8: Johansen cointegration and VECM price-discovery results for primary Kraken channels (no-FF sample).

Channel	Rank	k_Δ	Trace$_{r=0}$	α_{USD}	α_{other}	IS$_{\text{USD}}$ [0.69, 0.77]	Leader
BTC/USD vs BTC/USDC	0	2	7.71	—	—	—	undetermined
BTC/USD vs BTC/USDT	1	8	27.03	0.0079	0.0118	0.73	BTC/USD

95% critical value for trace r=0: 15.49. IS = Hasbrouck (1995) midpoint; bounds [0.69, 0.77].

Table 8 shows Kraken BTC/USD vs. BTC/USDT rejects no cointegration (rank = 1; stable at $k_\Delta \pm 1$). Hasbrouck (1995) IS bounds via both Cholesky orderings [7]: IS$_{\text{USD}} \in [0.69, 0.77]$, midpoint **0.73**—BTC/USD accounts for ≈73% of permanent price discovery. Bounds are tight (width 0.08), so leadership is not ordering-driven. Residual correlation 0.889 confirms high co-movement. For BTC/USD vs. BTC/USDC, no-FF does not reject rank = 0, consistent with long-run instability over this short window.

Figure 13 provides complementary visual evi-

dence via orthogonalized impulse-response functions from a bivariate VAR on 1-minute returns (AIC-selected 10 lags; Cholesky ordering: BTC/USD first, consistent with its IS leadership; N=26,548). A shock to BTC/USD transmits significantly to BTC/USDC (Panel C: ≈6 bps at impact, decaying in 3–4 minutes), while the reverse channel (Panel B) is indistinguishable from zero—the 95% CI envelops zero at all horizons. This asymmetry confirms that fiat-quoted prices lead stablecoin-quoted prices.

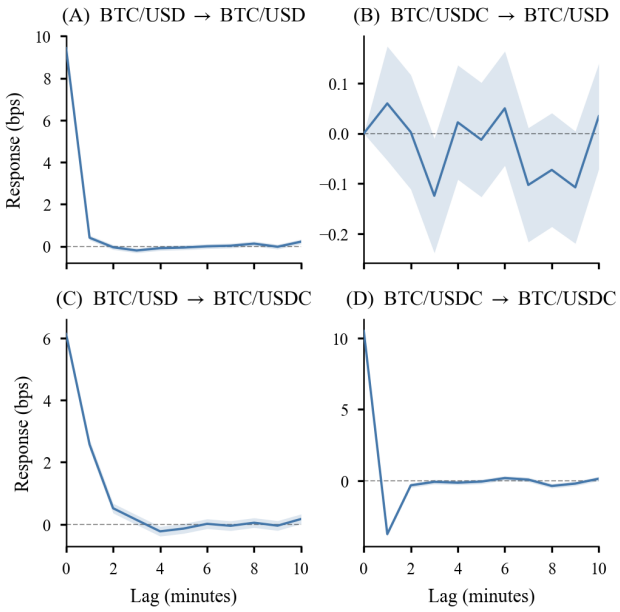

Figure 13: Orthogonalized IRF from VAR(10) on Kraken BTC/USD and BTC/USDC returns (bps). (B) USDC→USD: insignificant. (C) USD→USDC: significant (≈6 bps, decays in 3–4 min). Shaded: 95% asymptotic CI.

BH/FDR-corrected Granger tests on returns provide secondary evidence: BTC/USD → BTC/USDC is significant ($q < 0.001$) but the reverse is not after FDR correction ($q = 0.13$), consistent with fiat-to-stablecoin price adjustment. All other channels (USDT intra-exchange, cross-exchange) show bidirectional significance, consistent with active two-way arbitrage in more liquid markets.

3.5 Arbitrage after fees

Having established that price discovery resides in the fiat-quoted channel, a natural question is whether the documented dislocations offer exploitable arbitrage opportunities. We evaluate this using explicit trade construction—3-leg triangular for intra-exchange stablecoin channels, 2-leg prefunded buy/sell for cross-exchange—under two cost bounds:

$$\text{cost}_t^{\text{fee}} = n_{\text{legs}} \cdot f, \tag{7}$$

$$\text{cost}_t^{\text{fee+slip}} = n_{\text{legs}} \cdot f + \sum_{\ell=1}^{n_{\text{legs}}} \tfrac{1}{2}\,\text{RangeProxy}_{\ell,t}, \tag{8}$$

with $f = 5$ bps per taker leg.

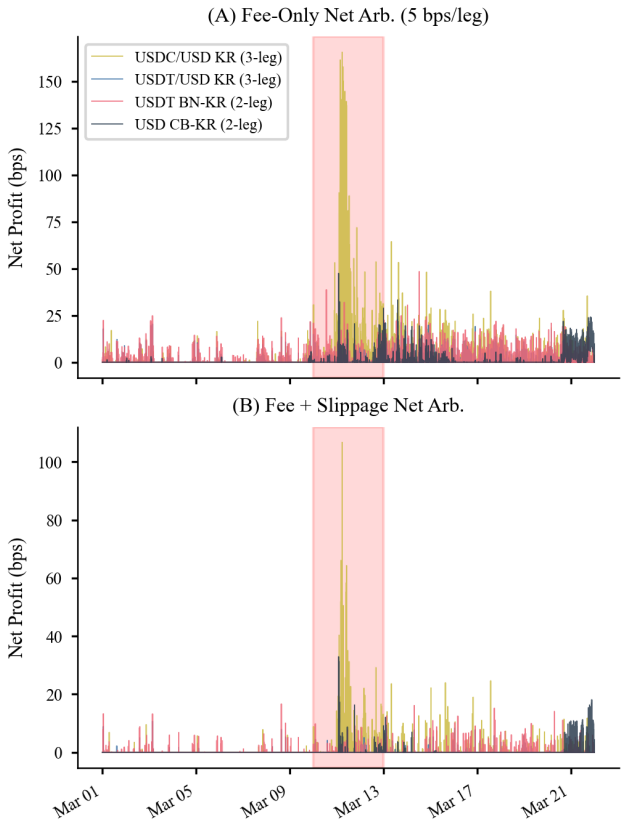

Figure 14: Net arbitrage profit (bps) under (A) fee-only costs (5 bps/leg) and (B) fee + range-based slippage. Crisis USDC profits compress sharply with slippage; cross-exchange profits are also attenuated.

Figure 14 shows the time series of net arbitrage profits under both cost specifications. Adding range-based slippage (Panel B) materially compresses profitability: USDC intra-exchange crisis profitability falls from 29.16% to 6.72% (net 4.83 to 0.64 bps); Coinbase–Kraken BTC/USD falls from 11.32% to 2.29%. Fee sensitivity: at f=3 bps (maker rebate), USDC crisis fee+slippage profitability rises to 14.0%; at f=10 bps (retail), it falls to 1.0%. The narrow post-cost opportunities are consistent with limited executable arbitrage under realistic frictions.

Table 9: Arbitrage profitability by channel/regime with 5 bps taker fees per leg (3-leg intra-exchange; 2-leg cross-exchange).

Channel	Regime	Cost Variant	%Profitable	AvgNetUncond (bps)
USDC/USD (Kraken)	Crisis	Fee-only	29.16	4.83
USDC/USD (Kraken)	Crisis	Fee+slip	6.72	0.64
USDC/USD (Kraken)	Post-SVB	Fee-only	9.38	0.56
USDC/USD (Kraken)	Post-SVB	Fee+slip	1.78	0.08
USDT/USD (Kraken)	Crisis	Fee-only	3.76	0.14
USDT/USD (Kraken)	Crisis	Fee+slip	0.37	0.01
Cross BTC/USD (CB–KR)	Crisis	Fee-only	11.32	0.72
Cross BTC/USD (CB–KR)	Crisis	Fee+slip	2.29	0.15
Cross BTC/USDT (BN–KR)	Crisis	Fee-only	11.85	0.49
Cross BTC/USDT (BN–KR)	Crisis	Fee+slip	1.81	0.05

4 Regulatory Interpretation

The preceding analysis identifies a clear failure chain: reserve-access risk drives peg deviation, peg deviation drives dispersion—both mechanically through D_t and via the contagion channel $\hat{\lambda}$ through B_t—and limited depth constrains arbitrage correction. We now ask whether the GENIUS Act's provisions target this chain. The results point to banking-layer reserve-access risk as the root cause. Circle held \$3.3B ($\approx$8%) of USDC reserves at SVB [3]; the $\sim 940\times$ half-life gap (bootstrap $p<10^{-4}$) localizes persistent dislocation to the peg layer, and the significant $\hat{\lambda}$ confirms that this dislocation propagates beyond the mechanical channel. Sustained recovery required emergency deposit guarantees on March 16 [5].

The GENIUS Act (July 2025) targets four dimensions of this chain; we map provisions to results as consistency checks [11]. (i) *Reserve composition*: Table 10 maps 25%–75% mitigation to implied D_t of $\approx$240–80 bps. (ii) *Timely redemption*: the 572-min peg half-life (vs. 1.06 pre-SVB) quantifies the five-day redemption freeze. (iii) *Bankruptcy super-priority*: targets P99 tail risk (81.4 vs. 13.6 bps). (iv) *Transparency*: traders learned of SVB exposure only from Circle's Friday-night disclosure, consistent with the USDC–USDT decorrelation (ρ=0.16).

Table 10: Illustrative GENIUS Act scenario range (not causal): assumed mitigation of the reserve lock-up shock mapped to implied crisis D_t means from observed moments. Tail benchmark: USDC crisis B_t P99 = 81.4 bps vs. pre-crisis 13.6 bps.

Scenario	Mitigation (%)	Implied D_t (bps)	Reduction (bps)	Policy mapping
Observed baseline	0	320.0	0.0	SVB/No GENIUS baseline
Low mitigation	25	240.0	80.0	Reserve composition + redemption + transparency
Moderate mitigation	50	160.0	159.9	Reserve composition + redemption + transparency
High mitigation	75	80.1	239.9	Reserve composition + redemption + transparency
Full-mitigation lower bd	100	0.1	319.9	Reserve composition + redemption + transparency

Persistent cross-exchange deviations (+2.18 bps) and the $26\times$ depth gap suggest reserve-layer reforms alone may not eliminate spatial fragmentation.

5 Conclusion

Four findings emerge. First, SVB-triggered fragmentation is attributable to stablecoin peg risk: D_t reaches $+320$ bps but B_t stays near $+5$ bps with sub-minute mean reversion; the $\sim 940\times$ half-life gap is statistically confirmed (sieve bootstrap 95% CI $[216,\ 1{,}898]$, $p<10^{-4}$), localizing persistent dislocation to reserve-access risk. Second, the contagion intensity model reveals that peg stress transmits to the adjusted basis beyond the mechanical channel: $\hat{\lambda}$ is significant during crisis ($p<0.001$) but not pre-SVB, with USDC and USDT transmitting in opposite directions consistent with depeg versus safe-haven dynamics. Third, the crisis is USDC-specific: excess kurtosis 12.9 ($6\times$ tail expansion), USDT safe-haven premium ($+58$ bps), and cross-stablecoin decorrelation ($0.44 \rightarrow 0.16$). Fourth, post-friction arbitrage narrows sharply (29% $\rightarrow$ 7%), with the $26\times$ depth gap limiting executable scale.

The GENIUS Act's provisions target precisely this failure chain. Because dislocation resides in the peg layer and propagates to the adjusted basis through the λ channel, reforms reducing reserve-access risk should compress not only D_t mechanically but also B_t through the contagion pathway. However, persistent cross-exchange deviations and depth asymmetry suggest additional market-structure measures may also be needed. A natural next step is a post-GENIUS replication study during the next stress event.

Interpretation boundaries. This evidence is window-specific (March 2023) and observational; $\hat{\lambda}$ measures Granger-predictive transmission, not structural causation. Policy passages are consistency mappings, not causal identification. Robustness: (i) HAC intervals preserve directional signs; (ii) no-FF filters preserve rank-order conclusions; (iii) $\hat{\lambda}$ is robust to no-FF and 5-minute sampling; (iv) cointegration is rank-stable for BTC/USD–USDT but not USDC; (v) Chow tests confirm the regime split.

Reproducibility. All exhibits are deterministic outputs of `run_all.py`; artifact-label integrity is verified by `src/04_tex_integrity_check.py`.

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